

CODING

THONNY IDE - An Integrated Development Environment (IDE) is a software application that enables programmers to efficiently develop code by consolidating essential tools (all the tools are in 1 place)—a code editor, debugger, and compiler—into a single graphical interface. IDEs boost productivity by automating tasks, highlighting syntax, and offering code suggestions.

🔗 WEEK 1 PYTHON LESSON PLAN (BEGINNER FRIENDLY)

📄 LESSON 1: What is Programming?

Lets Start with real-world thinking

🍵 Example: Making Tea

“How do you make tea?”

Then break it into steps:

1. Boil water
2. Add tea powder
3. Add milk
4. Add sugar
5. Stir well
6. Serve

Now explain:

🔗 This is called an **algorithm** what does that mean?

An algorithm is a precise, step-by-step set of instructions or rules followed to solve a specific problem or perform a task. They operate on an input-process-output model, taking data, applying logical steps, and producing a result.

Now connect to computers

Key idea:

- A **program** = a set of instructions for a computer
 - **Coding** = writing those instructions in a language the computer understands
 - Computers are **very literal**
 - They do EXACTLY what you say
 - No guessing, no common sense
-

Funny example (important!)

If you say:

“Make tea”

A computer will not know what to do 😊

You must break it into exact steps like a recipe.

Mini Activity

Ask student to write instructions for:

“Opening Google Chrome and searching for ‘cats’”

Then show how detailed instructions must be.

Explain: computers need **very exact steps**

✦✦ Better version:

1. Turn on the computer
2. Click on the **Google Chrome** icon
3. Wait for the browser to open
4. Click on the search bar at the top
5. Type **"cats"**
6. Press **Enter**
7. Look at the results

Missing details = computer gets confused
Every small step matters

This is exactly how programming works

❑ LESSON 2: First Look at Python (IDE SETUP)

Open **THONNY IDE**

Explain:

- This is where we write and run code
 - It is like a "chat with the computer"
-

💬 ***First command: print()***

✦✦ What it does:

It displays text on the screen.

Code:

```
print("Hello World") (show that python is case sensitive)
```

Explain:

- `print` = show something, it's a key word

Python keywords are reserved words that form the foundation of the language's syntax and structure.

" " → text (string) (It is used to store anything made of characters, like: words ,sentences ,letters. Simply said A **string is anything inside quotes** , Always written inside " " or ' ' Python treats it as **text**, not math ,Even numbers become strings if it is inside quotes:)

```
age = "20" # this is a string
```

```
age = 20 # this is a number (integer)
```

- Computer just displays it

Practice:

Ask student to try:

```
print("My name is ...")
print("I am learning Python")
```

LESSON 3: Variables (Storing Information)

Explain simply:

A **variable** is like a box that stores information.

Real-life analogy:

- Box labeled "Name" → stores your name
- Box labeled "Age" → stores your age

Code:

```
name = "Tamasha"
a=5
```

- = means "store"
 - Python remembers values

```
b=8
```

```
print(name)
print(a)
print(b)
print(a+b)
```

Practice:

Create variables to store:

- Your name
 - Your age
 - favorite food
 - Hometown and display them
-

❑ LESSON 4: Taking an Input (Talking to the User)

Concept:

Now program can “ask questions”

Code:

```
name = input("What is your name? ")
print("Hello", name)
```

Explain:

- `input()` = takes user input
 - Program becomes interactive
-

🔗 Practice:

- Ask school
- Ask favorite color

```
name = input("Enter your name: ")
```

```
age = input("Enter your age: ")
```

```
print("Name:", name)
```

```
print("Age:", age)
```

```
print("Welcome to Python!")
```

📦 LESSON 5: Simple Logic (VERY LIGHT INTRO)

Only introduce idea (not deep yet)

Concept:

Computers can make decisions, but only using rules we give them.

Ask:

“How does a driver know when to stop or go at a traffic light?”

Answer:

A driver follows rules:

- If the light is **green** → go
- If the light is **red** → stop
- If the light is **yellow** → slow down

How to know whether someone is an adult or not?

- If age is 18 or above → adult
- If below 18 → child/minor

In Python, we write this decision using:

```
Code
light = "red"
if light == "green":
    print("Go")
else:
    print("Stop")
```

☞ if statement

Example:

```
age = 18

if age >= 18:
    print("You are an adult")
```

- `age = 18` → storing a value
- `if` → checking a condition (computer checks true or false)
- `>=` → “greater than or equal to”
- `:` (colon) this is part of python syntax (syntax punctuation) → means “A block of code is coming next”
- indentation (the blank space between a margin and the start of a text line, This is also part of Python syntax rules.) It tells Python:

“These lines belong inside the block”

→ here it tells Python what belongs inside the if

Ex: `if age >= 18:`
`print("You are an adult")`

Everything indented under `if` is grouped together.

- `:` to start a block
- indentation to define the block

Python is very strict about indentation (`:`) – (show using age coding)

Activity

```
Time=10.00
if time<12:
    print("Good Morning")
```

Add ELSE (VERY IMPORTANT NEXT STEP)

Now show decision “otherwise”:

```
age = 15

if age >= 18:
    print("You are an adult")
```

```
else:  
    print("You are a minor")
```

💡 *Explain:*

- `if` = first condition
- `else` = everything else

☞ Only ONE will run

Mini activity

☞ Write a program:

Create a variable called marks and assign any value you want between 0-100

- If marks are 75 or above → print “You’ve Passed”
- Else → print “You’ve Failed”